

Introduction to Linux Command

Display calendar

```
$ cal
```

File and Directory Management

List file and Directory

```
$ ls
```

List file and Directory with permission

```
$ ls -l or ls -lh
```

List hidden file and Directory

Hidden file represented by (.)

```
$ ls -a
```

List file and Directory with metta data

```
$ ls *.doc
```

List directories but also subdirectories

```
$ ls -R
```

Creating Directories

- Command : `mkdir`
- syntax: `mkdir` directoryname

- sample: `home@VirtualBox:~$ mkdir mydirectory`

To rename file and Directory

- Command: `mv`
- Syntax: `mv <file with existing name> <file with new name>`
- Sample: `simeon@simeon-laptop:~/CSC$ mv mysecondwork mysecondwork2`

Check the history of command used

- Command: `history`
- Syntax and sample: `simeon@simeon-laptop:~$ history`

Editing new or existing file

`$ vi filename` eg `$ vi OSLevel`

press the key `i` to come into the edit mode.

Creating & Viewing Files

The 'cat' command is used to display text files. It can also be used for copying, combining and creating new text files. Let's see how it works.

To create a new file, use the command

1. `cat > filename`
2. Add content
3. Press 'ctrl + d' to return to command prompt.

↪ Create a File
guru99@VirtualBox:~\$ cat > sample1

↪ Enter Content
This is sample1

↪ Press Control + D to exit
guru99@VirtualBox:~\$

To view a file, use the command -

cat or cat file_name

Delete file

- Command : rm
- syntax: rm <filename>
- sample: simeon@simeon-laptop:~/BIT\$ rm myFirstwork

Removing Directories

- Command : rmdir
- syntax: rmdir mydirectory
- sample: home@VirtualBox:~\$ rmdir mydirectory

Rename file and Directory

- command: mv
- syntax: mv <file with existing name> <file with new name>
- Sample: simeon@simeon-laptop:~/CSC\$ mv mysecondwork mysecondwork2

Changing directory

Command: cd

Single dot (.) – This represents the current directory.

Double dot (..) – This represents the parent directory.

File Permission

Set Permission by using Symbolic mode

Add and Remove permission to user group and all

1. Use `ls -l =>` to check permission assigned
2. Eg add write permission to the group

Syntax:

Add

`chmod <groupName>-<permissionName> <fileName>`

`chmod g+w name of file` to be assigned permission

Remove

`chmod <groupName>-<permissionName> <fileName>`

`chmod g-w name of file` to be assigned permission

```

simeon@simeon-laptop:~$ ls -l
total 44
drwxr-xr-x 2 simeon simeon 4096 2018-10-03 15:39 BIT
drwxr-xr-x 2 simeon simeon 4096 2018-10-03 15:46 CSC
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Desktop
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Documents
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Downloads
-rw-r--r-- 1 simeon simeon 179 2018-10-02 06:31 examples.desktop
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Music
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Pictures
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Public
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Templates
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Videos
simeon@simeon-laptop:~$ chmod g+w BIT
simeon@simeon-laptop:~$ ls -l
total 44
drwxrwxr-x 2 simeon simeon 4096 2018-10-03 15:39 BIT
drwxr-xr-x 2 simeon simeon 4096 2018-10-03 15:46 CSC
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Desktop
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Documents
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Downloads
-rw-r--r-- 1 simeon simeon 179 2018-10-02 06:31 examples.desktop

```

To add permission to all the groups together

- **Syntax:** chmod a+<permissionName> <fileName>
- **Example:** chmod a+w file

To add permission to all the groups without typing a

- **Syntax:** chmod +<permissionName> <fileName>
- **Example:** chmod +w file

To set explicit permissions for different groups

- **Syntax:** `chmod <groupName>=<permissions> <fileName>`
- **Example:** `chmod u=rwx,g=rw,o=r BIT`

Setting Octal (NUMBERS) Permissions (Absolut Mode)

Octal permissions can also be set for the groups.

777 = `rwxrwxrwx`

765 = `rwxrw-r-x`

654 = `rw-r-xr—`

Eg: `chmod 777 BIT`

For example, to set **r** octal will be **4**, to set **w** octal will be **2**, to set **x** octal will be **1**.

```
simeon@simeon-laptop:~$ chmod 777 BIT
```

Filtering

The `grep` Command

Case study

```
drwx-wxr-- 2 simeon simeon 4096 2018-10-10 03:14 BCS
drwxrwxrwx 2 simeon simeon 4096 2018-10-03 15:39 BCI
drwxr-xr-x 2 simeon simeon 4096 2018-10-03 15:46 CSC
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Desktop
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Documents
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Downloads
-rw-r--r-- 1 simeon simeon 179 2018-10-02 06:31 examples.desktop
-rw-r--r-- 1 simeon simeon 5 2018-10-10 03:39 hello
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Music
-rw-r--r-- 1 simeon simeon 79 2018-11-05 10:15 myFirstFile
drwxr-xr-x 2 simeon simeon 4096 2018-11-05 10:10 OS_Weeke-end
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Pictures
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Public
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Templates
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Videos
simeon@simeon-laptop:~$ ls -l |grep "2018-10-10"
```

1. Select all file created on "2018-10-10"

```
ls -l |grep "2018-10-10"
```

Command and result

```
simeon@simeon-laptop:~$ ls -l |grep "2018-10-10"
drwx-wxr-- 2 simeon simeon 4096 2018-10-10 03:14 BCS
-rw-r--r-- 1 simeon simeon 5 2018-10-10 03:39 hello
```

2. Select all file created on "2018-10-02 06:37"

Command and result

```
simeon@simeon-laptop:~$ ls -l |grep "2018-10-02 06:37"
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Desktop
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Documents
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Downloads
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Music
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Pictures
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Public
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Templates
drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Videos
```

3. Select all file created on "2018-10-02 06:37" with corresponding lines **number**
Command and result **-n**

```
simeon@simeon-laptop:~$ ls -l |grep -n "2018-10-02 06:37"
5:drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Desktop
6:drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Documents
7:drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Downloads
10:drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Music
13:drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Pictures
14:drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Public
15:drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Templates
16:drwxr-xr-x 2 simeon simeon 4096 2018-10-02 06:37 Videos
```

4. Prints **all lines that do not match pattern**. If pattern are all file created on "2018-10-02 06:37"
Command and result **-v**

```
simeon@simeon-laptop:~$ ls -l |grep -v "2018-10-02 06:37"
total 60
drwx-wxr-- 2 simeon simeon 4096 2018-10-10 03:14 BCS
drwxrwxrwx 2 simeon simeon 4096 2018-10-03 15:39 BIT
drwxr-xr-x 2 simeon simeon 4096 2018-10-03 15:46 CSC
-rw-r--r-- 1 simeon simeon 179 2018-10-02 06:31 examples.desktop
-rw-r--r-- 1 simeon simeon 5 2018-10-10 03:39 hello
-rw-r--r-- 1 simeon simeon 79 2018-11-05 10:15 myFirstFile
drwxr-xr-x 2 simeon simeon 4096 2018-11-05 10:10 OS_Wekee-end
```

5. Prints **only the count of matching lines**. If pattern are all file created on "2018-10-02 06:37"
Command and result **-c**

```
simeon@simeon-laptop:~$ ls -l |grep -c "2018-10-02 06:37"  
8
```

User account Management

Command to tell you about the system's username.

Command: **whoami**

Sample:

```
simeon@simeon-laptop:~$ whoami  
simeon
```

Command to tell you about the system's users.

Command: **users**

Sample:

```
simeon@simeon-laptop:~$ users  
simeon simeon
```

Command for logout

Type command: **logout**

Command for exit.

Type **exit**

Command gives the information about the users logged on to the system.

Command: **who**

Sample:

```
simeon@simeon-laptop:~$ who
simeon  tty7      2018-10-04 14:33 (:0)
simeon  pts/0     2018-10-04 14:33 (:0.0)
simeon@simeon-laptop:~$
```

This command tells about the users who are logged in and what are they doing.

Command: **w**

```
simeon@simeon-laptop:~$ w
 14:45:00 up 12 min,  2 users,  load average: 0.19, 0.17, 0.17
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU WHAT
simeon    tty7     :0               14:33    12:28  9.40s  0.43s gnome-session
simeon    pts/0    :0.0             14:33    0.00s  0.33s  0.02s w
simeon@simeon-laptop:~$
```

Sample:

Linux User Management

changing password

```
$ passwd
$ passwd
Changing password for amrood
(current) Unix password:*****
New Unix password:*****
Retype new Unix password:*****
passwd: all authentication tokens updated successfully
$
```

If more than one person use a single system, then everyone may have their own user account. Here, it will be helpful to know the user account details.

User management includes everything from creating a user to deleting a user on your system. User management can be done in three ways on a Linux system.

1. **Graphical tools** are easy and suitable for new users, as it makes sure you'll not run into any trouble.
2. **Command line tools** includes commands like `useradd`, `userdel`, `passwd`, etc. These are mostly used by the server administrators.
3. Third and very rare tool is to **edit the local configuration files** directly using `vi`.

Add a New User

To add/create a new user, all you've to follow the command '`useradd`' or '`adduser`' or "`sudo adduser username if you are not root`" with 'username'. The 'username' is a user login name, that is used by user to login into the system.

For example, to add a new user called '`NDAYAMBAJE`', use the following command.

```
sudo useradd username
```

eg: `sudo useradd simeon`

Delete / Remove User Account

`userdel` userName
`userdel [options] userName`
`userdel -r username`

`userdel` command examples

To remove the user named NDAYAMBAJE or account named NDAYAMBAJE from the local Linux system / server / workstation, enter:
`userdel` NDAYAMBAJE